

Alvin Rindra Fazrie
Senior AI Architect | Senior Data Scientist | MLOps & LLMOps Lead
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PROFESSIONAL SUMMARY

Senior AI and data professional with 10+ years of experience designing, deploying, and scaling enterprise AI, ML, and GenAI systems in production. Strong track record in enterprise AI architecture, MLOps and LLMOps standards, optimization, and NLP. Nominated as a Global MLflow Ambassador and certified as a Databricks GenAI Engineer, Microsoft Azure Data Scientist, TensorFlow Developer, and Google Cloud Professional ML Engineer.

PROFESSIONAL EXPERIENCE

Lufthansa Industry Solutions - Hamburg, Germany
Senior AI & Data Consultant | July 2019 – Present

- Architected and deployed **production-grade AI & data solutions** on **Databricks, OpenShift, Google Cloud, and Microsoft Azure**.
- Led **MLOps & LLMOps Workmode** to define standards, reusable templates, and enterprise architecture guidance.
- Developed and deployed the **Operations Research (OR)/Optimization** solution to optimize the sequence of Aircraft MRO (Maintenance, Repair, Overhaul) tasks using **CP-SAT Solver**.
- Productionized **GenAI/RAG Systems** for finance, HR, sales, & procurements use cases.
- Developed NLP-based Document Analytics Solution - reducing MRO Engineering efforts by **15,000+ man-hours annually**
- Architected hybrid ml recommendation systems to **optimize flight and destination**.

Tech Stack: **GenAI - RAG** (langchain, Ollama, Elasticsearch, Milvus, Atlas), **DL/ML** (TensorFlow 2.x, PyTorch, Sklearn, XGBoost, Catboost), **OR** (OR-Tools, HiGHs, CBC), **NLP** (gensim, spaCy, transformers), **Azure** (AKS, Azure ML SDK v2, Synapse, OpenAI), **Databricks** (Mosaic AI, MLflow, Asset Bundles, PySpark, Unity Catalog), **GCP** (AutoML, BigQuery, KubeFlow, VertexAI, TFX), **ML/LLMOps** (MLflow, Airflow, PySpark, Docker, Kubernetes, Helm)

DTSense Academy - Remote, Online

Lead GenAI & Data Instructor | Feb 2022 – Present

- Led and mentored data science team in various AI & data topics.
- Designed and taught end-to-end AI curricula covering **GenerativeAI - Large Language Model and Retrieval Augmented Generation**, AI & Machine Learning: - Supervised Learning (Vector-based: SVM & kNN; Probabilistic-based: Naive Bayes; Tree-based: Decision Tree, Deep Learning)
- Supervised hands-on projects, including Image Classification, Text Classification, Sentiment Analysis, and Multimodal Deep Learning models (text and audio).
- Participated in several Data Hackathons. Built Time Series (fbprophet, XGBoost) and Natural Language Processing (BERT) models.

Tech Stack: Python, pandas, scikit-learn, TensorFlow, MLflow, langchain, ollama, groq, fastapi, Looker Studio, fbprophet, XGBoost

Eraneos Analytics (formerly Ginkgo Analytics) - Hamburg, Germany

Data Scientist | March 2019 – June 2019

- Developed Analytics Platform using ELK Stack (Elasticsearch, Logstash, Kibana + Filebeat)
- Implemented Anomaly Detection with Machine Learning Jobs (Single Metric and Multi-Metric), and Time-series Analysis with Visual Builder.

- Participated in **Airbus AI Challenges**: achieving high-performance anomaly detection for Airbus Helicopters Accelerometer Data. Recall: 0.94, Precision: 0.99. ML Algorithms: OneClassSVM, K-Means, KNearestNeighbour
- Built Product Classifiers with Deep Learning algorithms (LSTM, CNN) in AWS Platform.

Tech stack: TensorFlow, scikit-learn, AWS (Sagemaker, Redshift, Quicksight, Lambda), Azure (SSIS, Azure Data Factory, Power BI)

University of Hamburg - Hamburg, Germany

Research Assistant - Language Technology Group | Sep 2017 – Aug 2018

- Developed AI-driven tools for investigative journalism and data-driven approach, including **new/s/leak** 2.0 (Network of Searchable leaks), Science and Data-Driven Journalism tool <http://www.newsleak.io/>, and **autolinks** (automatic proactive researching) - <https://uhh-lt.github.io/autolinks>, under the supervision of Prof. Chris Biemann.

Tech stack: Python, Scala Play, AngularJS, NodeJS, Elasticsearch, Docker, PostgreSQL, NLP components: UIMA, cTAKES, Polyglot-NER.

HAUSGOLD, xrails, quantilope - Hamburg, Germany

Full-Stack Developer | Feb 2015 – Jan 2018

- Built analytical dashboards and market research platforms across startups. Delivered full-stack solutions with modern frontend and backend technologies in agile environments.

Tech stack: Frontend: ReactJS, E2E Test: NightwatchJS. Backend: MeteorJS, RoR, PostgreSQL, Docker

EDUCATION

UNIVERSITY OF HAMBURG - Hamburg, Germany

M.Sc. Master of Intelligent Adaptive Systems | Oct 2014 – Dec 2018

- Focus areas: Artificial Intelligence, Machine Learning, Neural Networks / Deep Learning
- Research projects in Reinforcement Learning and Natural Language Processing

UNIVERSITY OF DUISBURG-ESSEN - North Rhine-Westphalia, Germany

SAP ERP & Business Intelligence Program | Nov 2014 – Mar 2017

- Completed SAP modules in Enterprise Resource Planning, Business Intelligence, and Advanced Business Application Programming.

UIN Jakarta - Jakarta, Indonesia

B.Sc. Computer Science | Sep 2009 – Oct 2012

- Focus areas: Software Engineering and Artificial Intelligence.
- 1-year Study Abroad in IIUM, Malaysia. 3 months working as an Associate System/BI Developer.

PROFESSIONAL CERTIFICATIONS

Global MLflow Ambassador | Dec 2025 - Jan 2027

https://alvinrindra.github.io/assets/docs/Alvin's_Certificates.pdf, <https://mlflow.org/ambassadors/>

Databricks Generative AI Engineer Associate | Nov 2025 - Nov 2027

<https://credentials.databricks.com/e543c1b3-f3b2-4291-8e09-f7ad8ee3364d#acc.METjvhvBI>

Skills: MLOps, LLMOps, Mosaic AI VectorSearch, LLM Fine-tuning, Retrieval Augmented Generation

Azure Data Scientist Associate | July 2023 – Aug 2027

Microsoft Certification: <https://learn.microsoft.com/en-us/users/alvinrindra/credentials/eeca8947aecbb75a>

Skills: AzureML, AutoML, MLDesigner, MLflow, MLOps, Azure Pipelines, GitHub Actions, ResponsibleAI, Databricks

Professional Machine Learning Engineer | April 2022 – April 2024

Google Cloud Certification: <https://www.credential.net/profile/alvinrindrafazrie965276/wallet>

Skills: TensorFlow2.x, TFX, KubeFlow, VertexAI, BigQuery, Apache Beam, AutoML, Natural Language API, Video Intelligence API, Dataproc, Dataflow, MLOps

TensorFlow Developer | July 2020 – July 2023

Google/TensorFlow Certification: <https://www.credential.net/profile/alvinrindrafazrie965276/wallet>

Skills: TensorFlow2.x, Supervised Learning (Classification & Regression), NLP, Computer Vision, Time-Series

Production Machine Learning Systems; MLOps; MLPipelines | Jan 2022

Coursera Certificates: <https://coursera.org/verify/BQ2QJPZ7KXX>,

<https://coursera.org/verify/professional-cert/K6N69EUFWKBP>

RESEARCH PUBLICATIONS

Action Selection Methods in a Robotic Reinforcement Learning Scenario. Cruz, F.; Wüppen, P.; Fazrie, A.; Weber, C.; Wermter, S. - Nov 2018

5th IEEE Latin American Conference on Computational Intelligence, LA-CCI. Tech stack: Python, NumPy, matplotlib. Link: <https://ieeexplore.ieee.org/document/8625243>.

Agent-advising Approaches in an Interactive Reinforcement Learning Scenario. Cruz, F.; Wüppen, P.; Magg, S.; Fazrie, A.; Wermter, S. - Sep 2017

2017 IEEE International Conference on Development and Learning and Epigenetic Robotics (ICDL-EpiRob). Tech stack: Python, NumPy, matplotlib. Link: <https://ieeexplore.ieee.org/document/8329809>.

SKILLS & INTERESTS

Technical Skills: Python3.x, **AI/Data Science** (TensorFlow2.x, PyTorch, scikit-learn, mlxtend, statsmodels, fbprophet, OR-Tools), **Data Engineering** (PySpark, AirFlow, Azure Synapse), **MLOps & Platforms** (VertexAI, TFX, AzureML, Databricks, MLflow), **Data Visualization** (Dash & plotly, streamlit, Looker Studio, Amazon Quicksight), **Backend** (RestAPI, FastAPI, Flask), **AI/Data Governance** (MLflow, Unity Catalog, Purview), **ResponsibleAI** (fairlearn, AI scorecard, interpret, SHAP), **GenAI/AgenticAI** (langchain, OpenAI, Ollama, semantic kernel, huggingface), **Computer Vision** (OpenCV, skimage), **Logs and Monitoring** (Prometheus + Grafana, Elasticsearch + logstash + Kibana)

Interpersonal: Communication, Emotional Intelligence, Teamwork, Attention to detail, Problem-Solving, Self-management, Leadership

Languages: English (Full professional proficiency), German (Working Proficiency), Indonesian (Native), Malay (Fluent)

Interests: Data Science, ML/Data Engineering, AI System Architecture, Mentoring, Boxing, Swimming, Organizations, Community & Charity Work